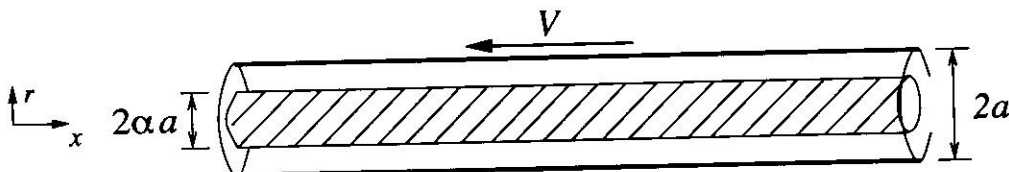


VF3

A fluid of density ρ and viscosity μ is confined between two infinitely long coaxial cylinders of radii a and αa , with $\alpha < 1$, aligned in the x direction. The fluid is forced to move steadily due to the presence of a known reduced pressure gradient $P_l = -\partial P / \partial x$ and also due to the backward motion of the outer cylinder, with velocity V .

1. Determine the fluid velocity $v_x(r)$.
2. Obtain the power needed to move the outer cylinder.
3. Calculate the value of V for which the volume flux in the pipe is identically zero.



$$0 = P_l + \frac{\mu}{r} \frac{d}{dr} \left(r \frac{dv_x}{dr} \right) \quad \left\{ \begin{array}{l} r=a: v_x = -V \\ r=\alpha a: v_x = 0 \end{array} \right\}$$

INTEGRATING TWICE $\rightarrow v_x = -\frac{P_l}{4\mu} \frac{r^2}{4} + C_1 \ln r + C_2 = -\frac{P_l}{4\mu} [r^2 - (\alpha a)^2] + \left[V + \frac{P_l (1-\alpha^2)a^2}{4\mu} \right] \frac{\ln(r/\alpha a)}{\ln(\alpha^{-1})}$

$$\tau_p = -\mu \left. \frac{dv_x}{dr} \right|_{r=a} = \frac{P_l a}{2} \left[1 - \frac{1-\alpha^2}{2 \ln(\alpha^{-1})} \right] + \frac{\mu V}{a \ln(\alpha^{-1})}, \quad W = \tau_p V 2\pi a$$

$$Q = \int_{\alpha a}^a 2\pi r v_x dr = 0$$

$$\frac{V}{\ln(\alpha^{-1})} \int_{\alpha a}^a r \ln(r/\alpha a) dr = \frac{P_l}{4\mu} \int_{\alpha a}^a \left[(1-\alpha^2) a^2 \frac{\ln(r/\alpha a)}{\ln(\alpha^{-1})} - r^2 + (\alpha a)^2 \right] r dr$$